The **largest** among them is the **Pox virus** measuring about 300nm. The **smallest** virus is **Parvo virus** measuring about 20 nm.

Structure and Shape

The virion consists of nucleic acid surrounded by a protein coat, **the capsid**. The capsid with the enclosed **nucleic acids** is known as the nucleo capsid. The capsid is composed of a large number of **capsomers**. The functions of the capsid are to **protect** the **nucleic acid** from the deleterious agents and also to introduce viral genome into host cells by adsorbing readily to cell surfaces (Figure 10.1). Two kinds of symmetry encountered in the virus are **icosahedral** (cubical) and **helical**. Virions may be **enveloped or non enveloped** (naked). The envelope or outer covering of viruses is derived from the host cell membrane when the progeny virus is released by budding. The envelope is **lipoprotein** in nature. The lipid is of host cell origin while the protein is virus coded. **Protein subunits** may be seen as **projecting spikes** on the surface of the envelope and are known as Peplomers.

Overall shape of the virus particle varies; mostly animal viruses are roughly spherical. Some are irregular and pleomorphic. The rabies virus is **bullet shaped**, Ebola virus is **filamentous** and pox viruses are **brick shaped**.

Chemical Properties

Viral protein determines the **antigenic specificity** of the virus. Some viruses contain small amounts of carbohydrates. Most Viruses do not possess any enzymes but **retro virus** has a unique enzyme, such

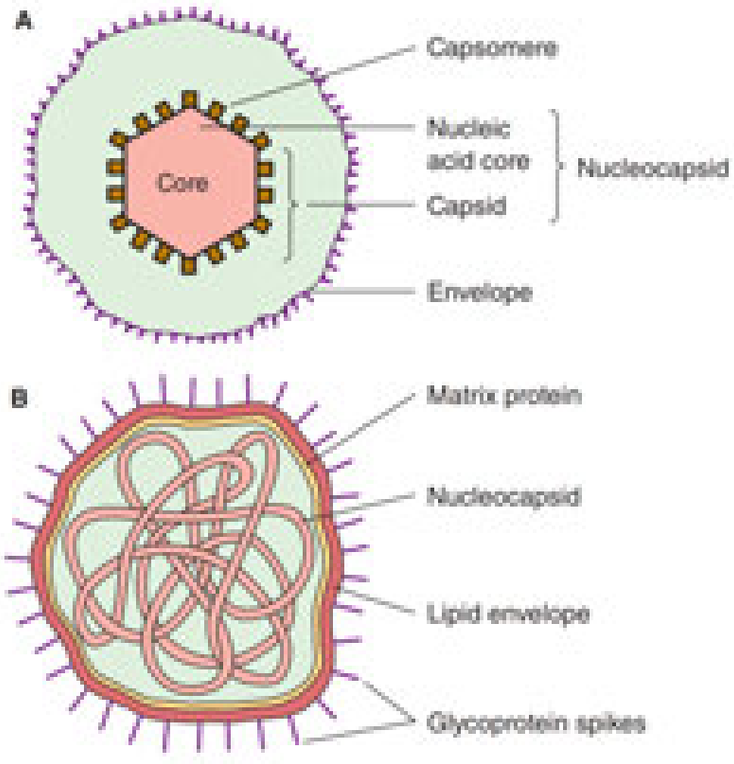


Figure 10.1: Structure of Virus

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